



AUTHOR PREDICTION CHATBOT USING MACHINE LEARNING

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ST10345096 Postgraduate Diploma in Data Analytics



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Introduction

As a data analyst for a large book publisher, the task was to create a fun tool that allows fans to interact with books and authors. The specific task was to develop a bot that could guess an author's identity based on their writing style.

The project used the Standardized Project Gutenberg Corpus (SPGC). Ten English-language authors were selected, and their books were cleaned, divided into passages and analysed to identify differences in writing style. Various text features and machine learning models were tested to determine which approach was most suitable.

The final solution was an author prediction chatbot. It accepts a passage of text and returns the three authors that the model considers most likely, together with their confidence scores.

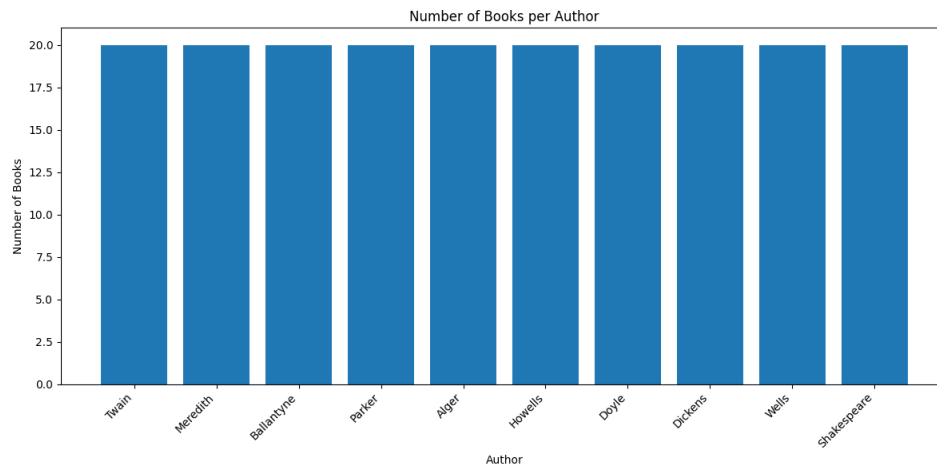
Dataset Evaluation and Suitability

The Standardized Project Gutenberg Corpus (SPGC) was selected because it provides a large collection of digitised books that can be used for authorship analysis. The dataset contains information such as book titles, authors and text, making it suitable for creating labelled training data.

The analysis was limited to English-language books and focused on ten selected authors. Authors with sufficient books were considered, while unknown, anonymous and multiple-author entries were removed. Up to 20 books per selected author were used to provide enough text for the analysis.

The selected books were divided into 500-word passages, while passages with fewer than 300 words were excluded. This produced many labelled text samples, with each passage linked to its original author.

The dataset was suitable for the project because authorship prediction depends on having enough text from multiple authors to identify differences in writing style. However, the dataset also has limitations, including differences in the number and length of books available for each author and the possibility that passages from the same book may occur in both the training and testing data.



Why this dataset was suitable

- Large amount of literary text.
- Clear author labels.
- Multiple books available per author.
- Suitable for text classification.
- Provides enough data to create training and testing passages.

Data Cleaning and Preparation

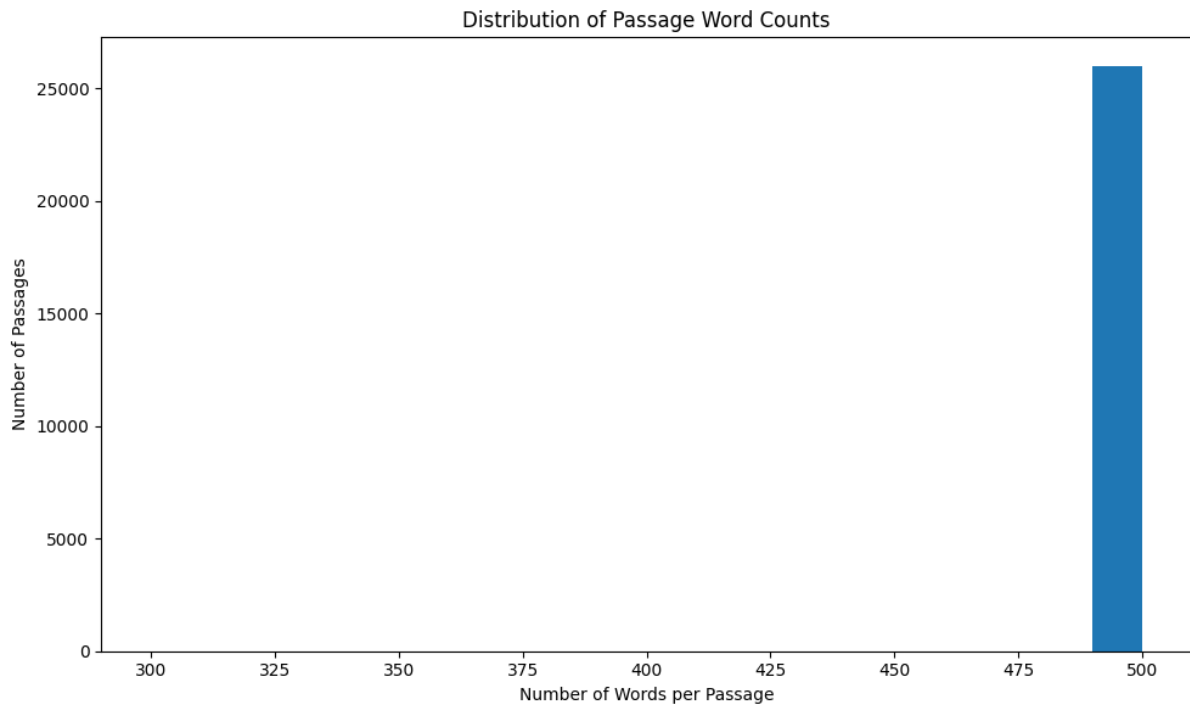
The raw SPGC data could not be used directly because it contained records that were not suitable for authorship classification. The data was therefore cleaned before modelling.

First, only English-language records were retained. Authors listed as Unknown or Anonymous were removed, together with records containing multiple authors, editors, translators or illustrators. This ensured that each text could be linked to a single author.

The ten selected authors were then used to identify suitable books. The downloaded book texts were cleaned by removing unnecessary Gutenberg text and retaining the actual book content.

Each book was divided into 500-word passages. Passages containing fewer than 300 words were removed because they provided less information about an author's writing style. Each remaining passage was assigned a unique record ID, book ID and author label.

The resulting dataset was then prepared for machine learning by converting the text into numerical features. The data was divided into training and testing sets so that the models could be evaluated on data that was not used during training.



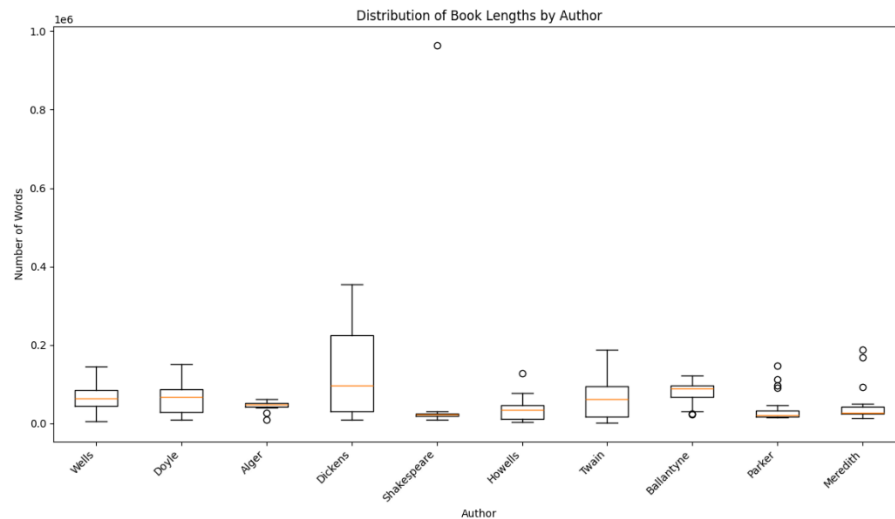
This preparation produced a consistent dataset in which each observation represented a passage of text linked to one of the ten selected authors.

Exploratory Data Analysis

EDA was conducted to understand the structure of the dataset and identify differences in writing characteristics between the selected authors. The analysis focused on both the size of the data and measurable characteristics of the text.

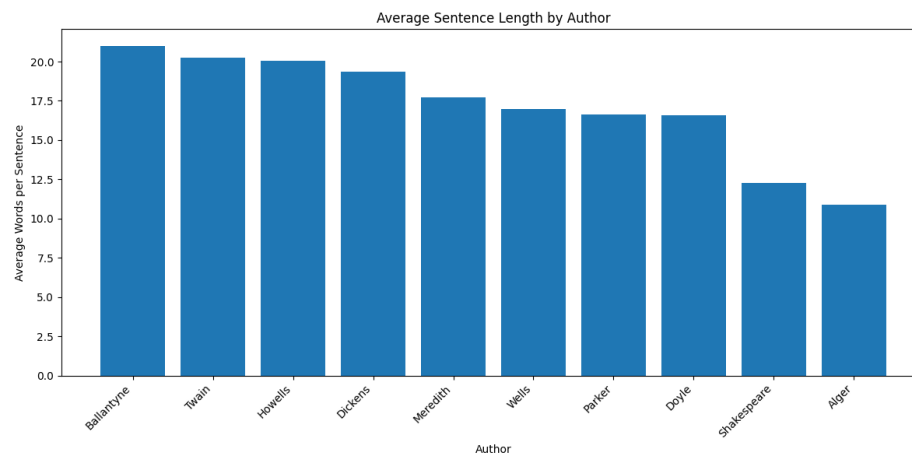
Dataset Structure

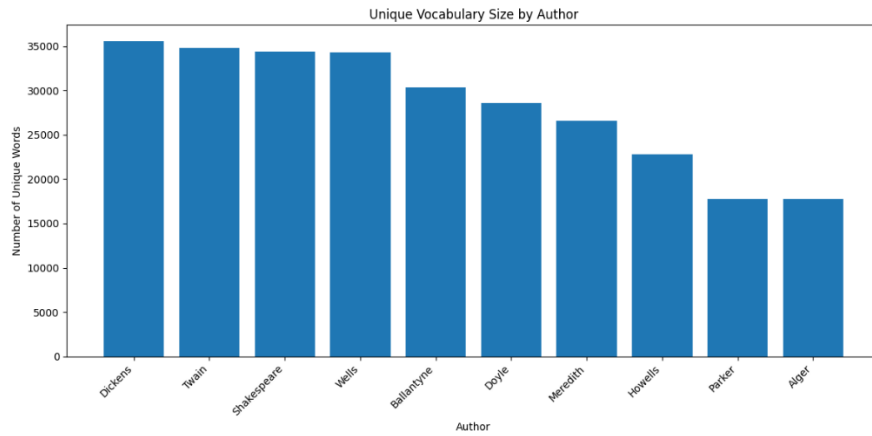
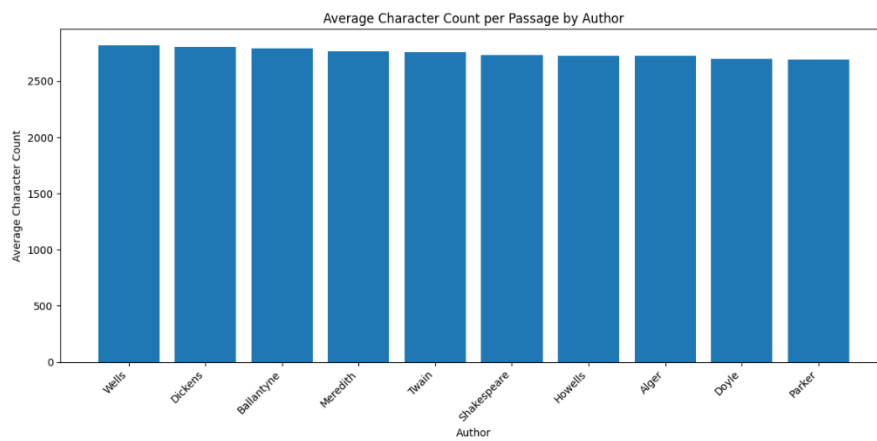
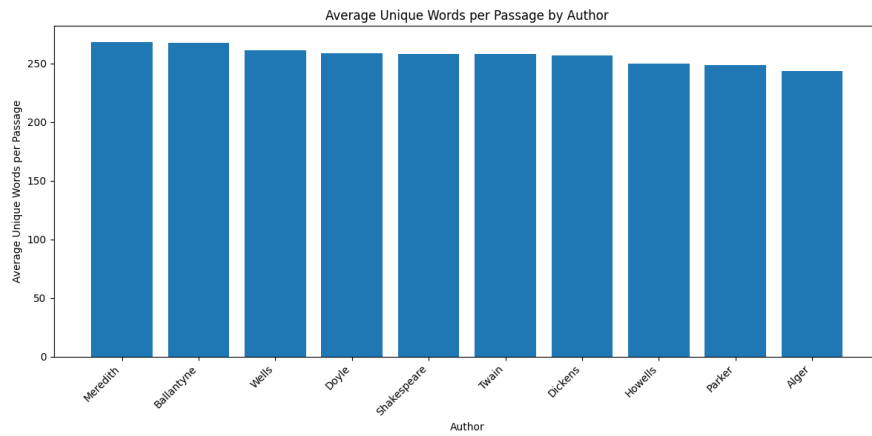
The number of books and passages was first examined for each author. This helped identify differences in the amount of available data and whether the dataset was reasonably balanced. The analysis also examined book and passage lengths.

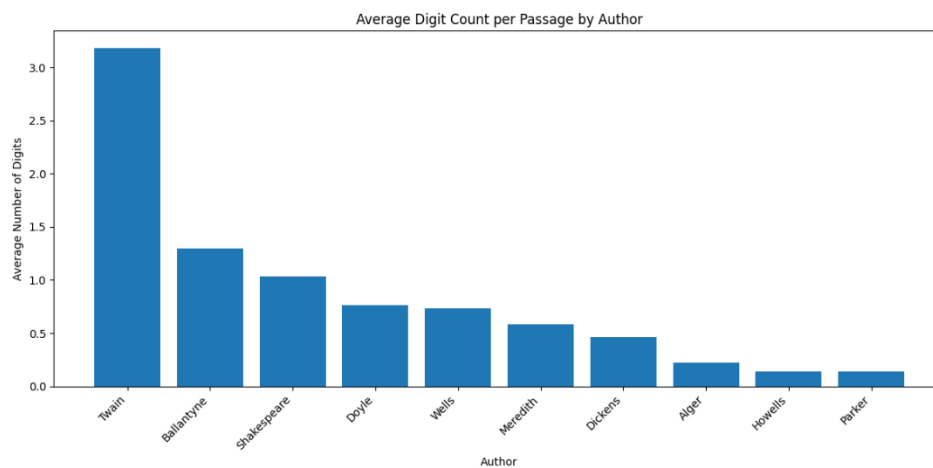
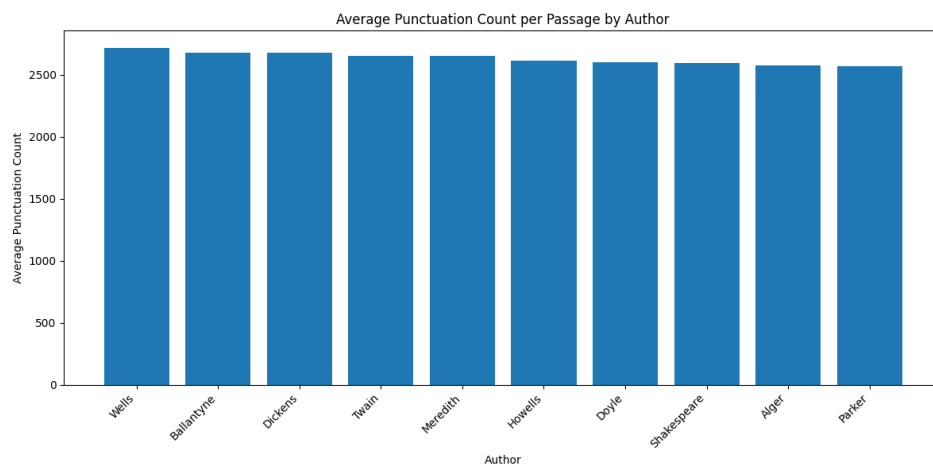
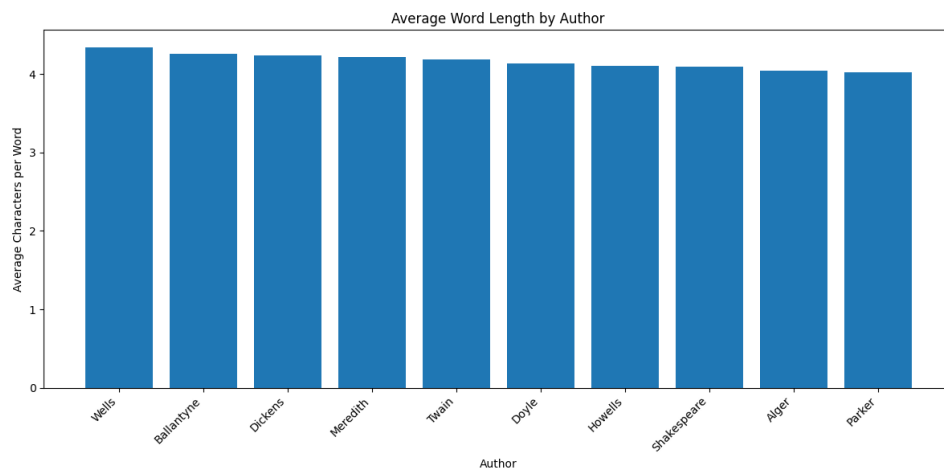


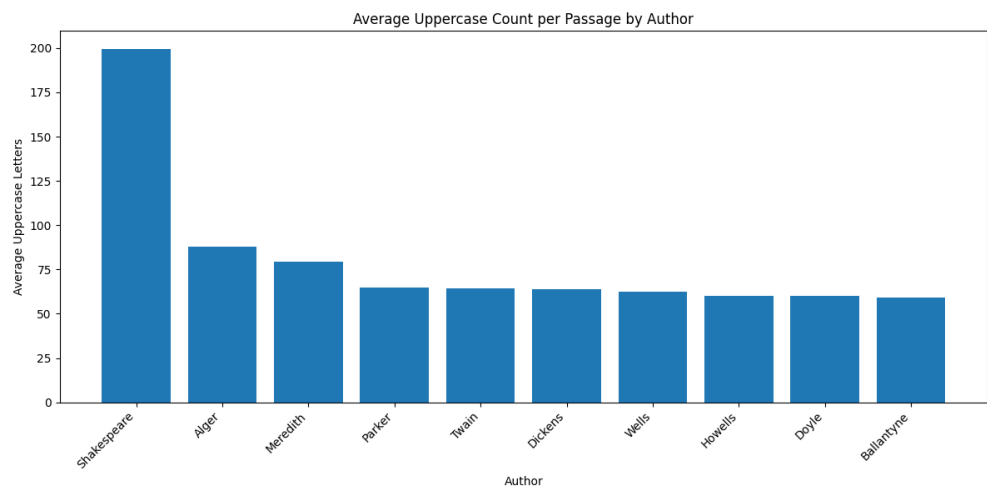
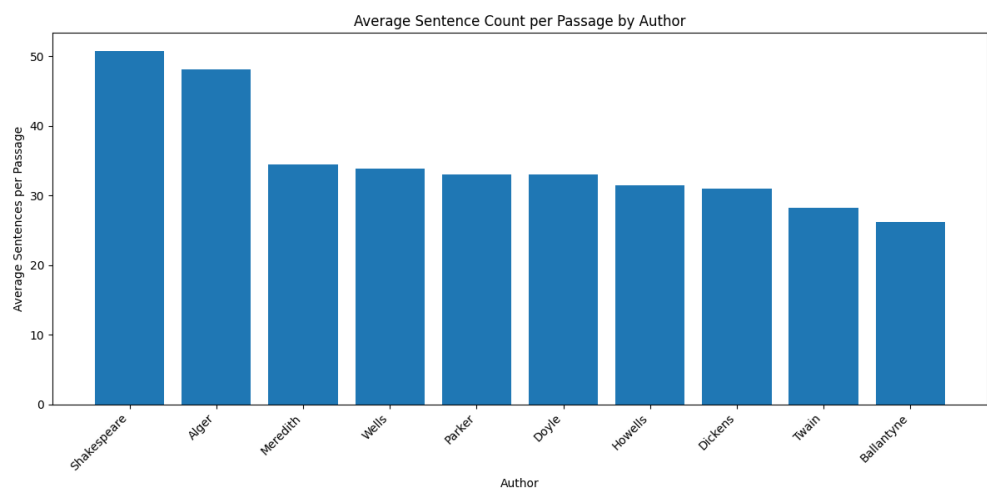
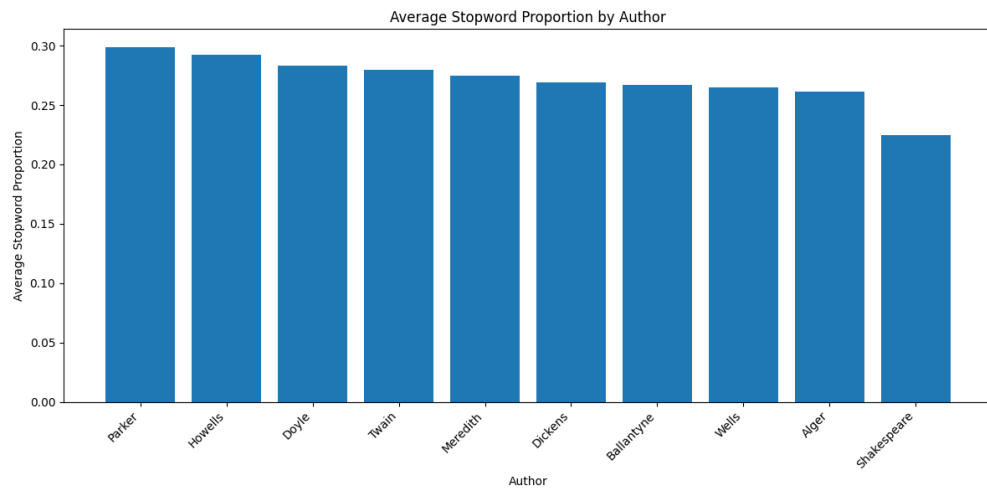
Writing Characteristics

Several text characteristics were calculated for each author. These included sentence length, vocabulary size, word length, punctuation, digits, stopwords and uppercase characters.



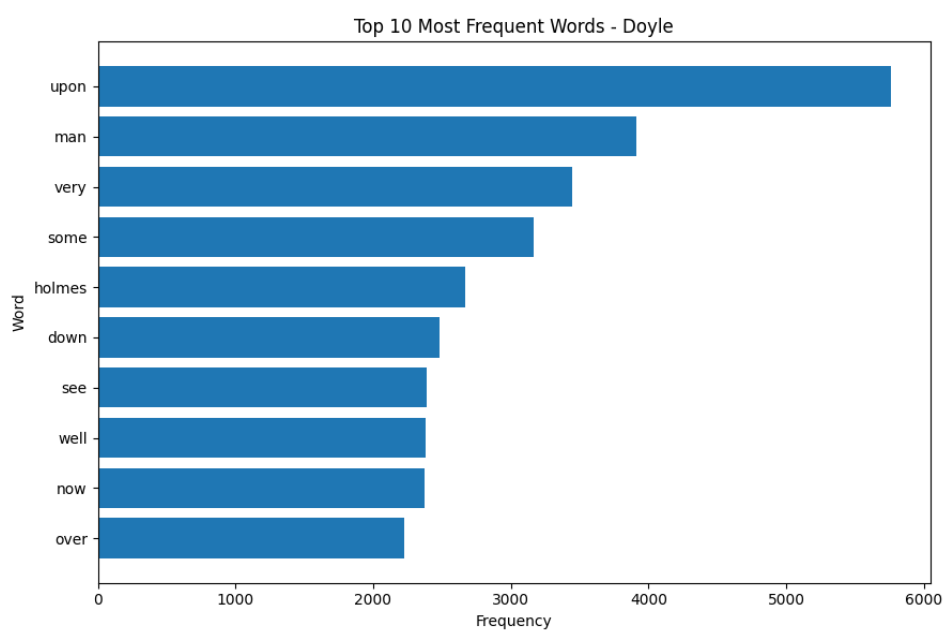
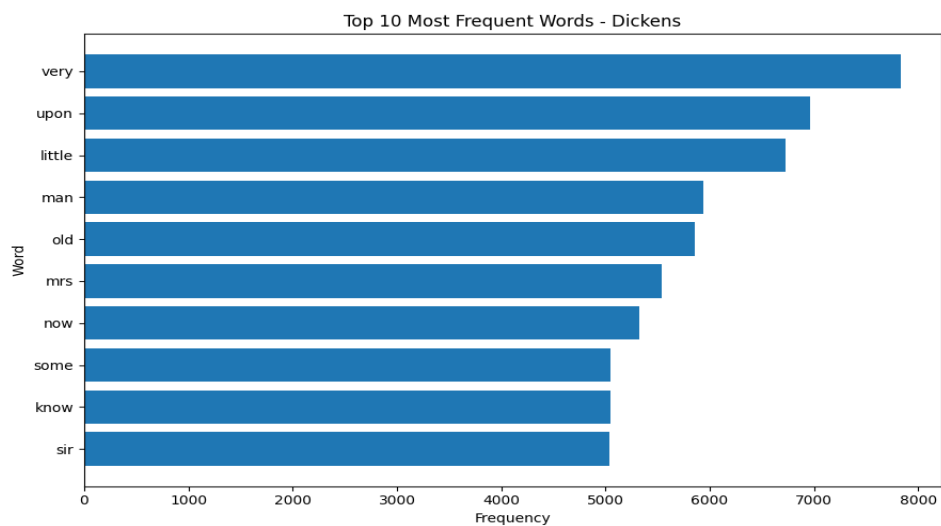
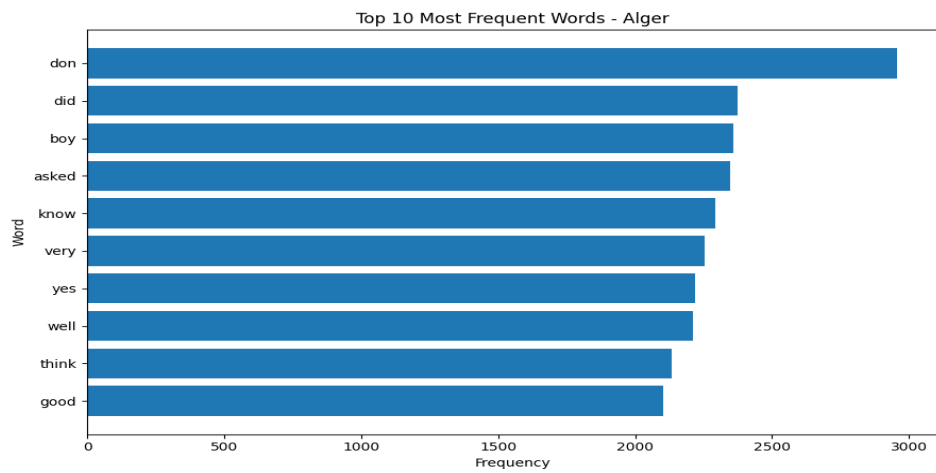


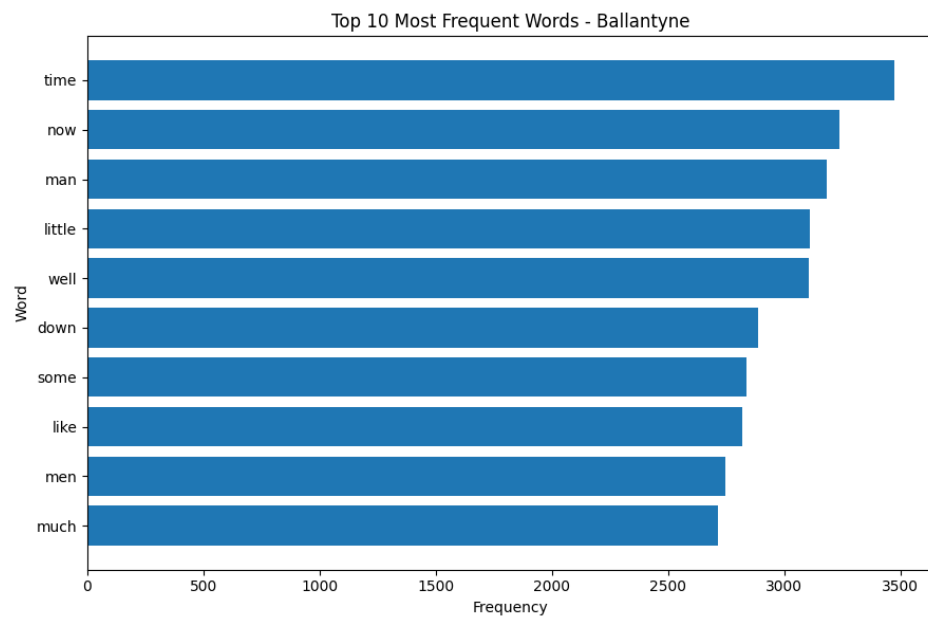
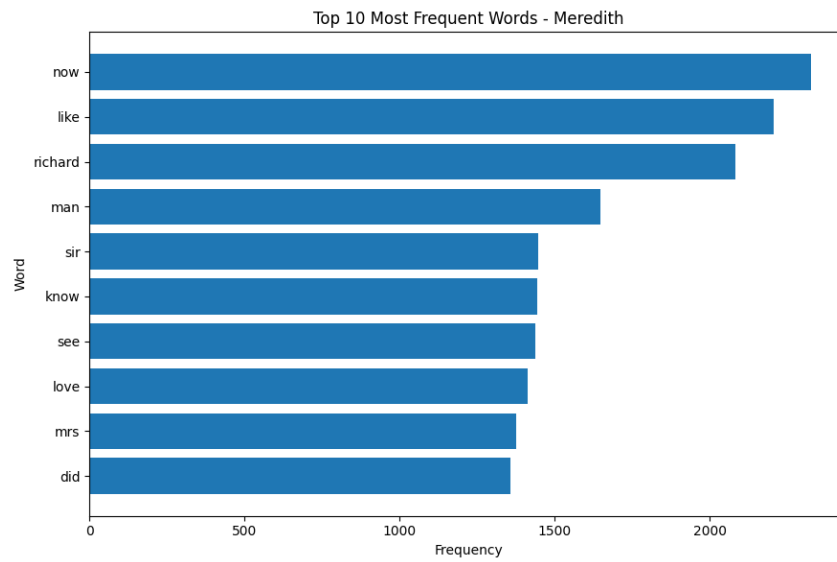
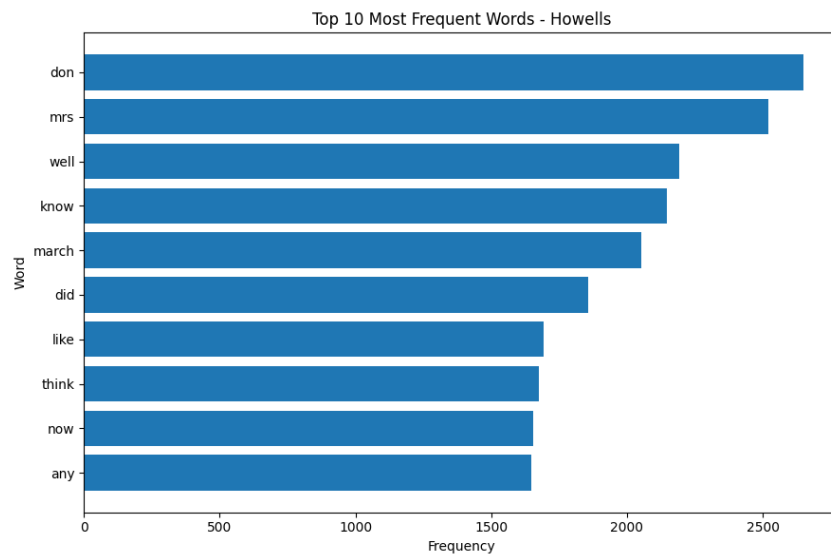


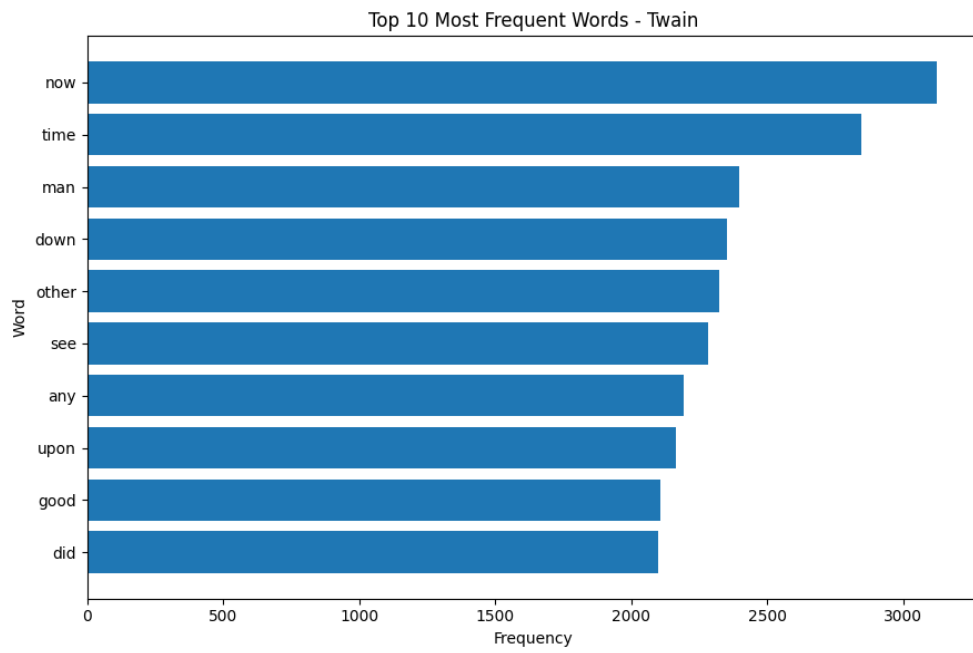
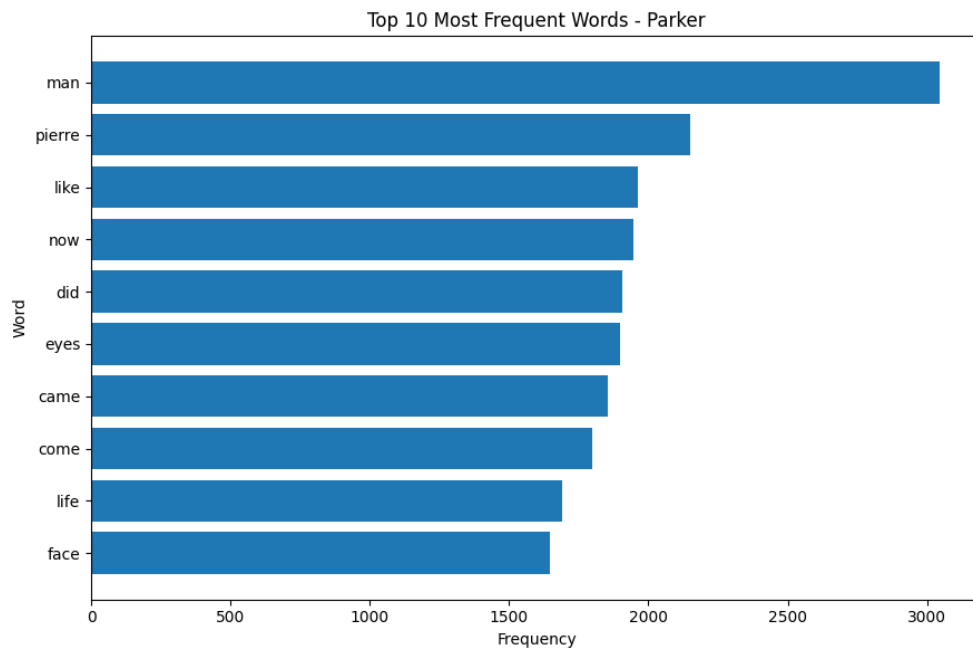


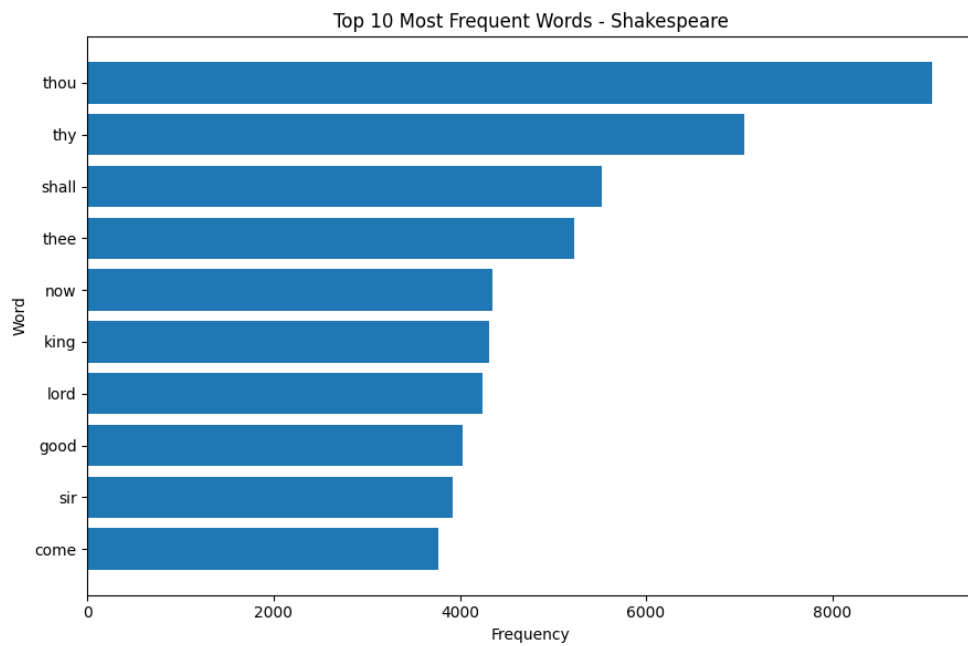
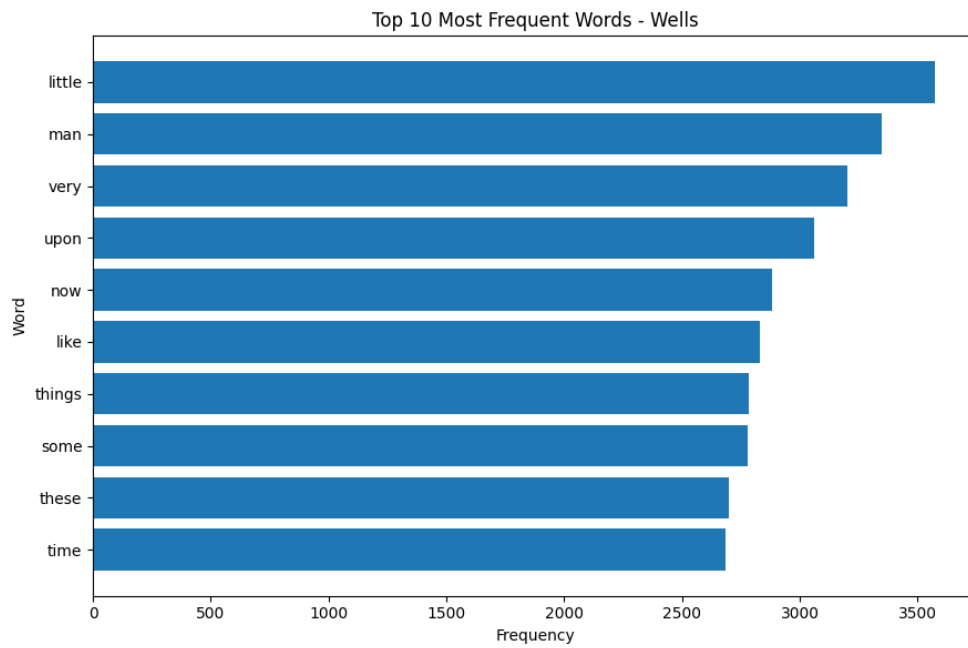
Vocabulary and Word Usage

The most frequently used words were also examined for each author.

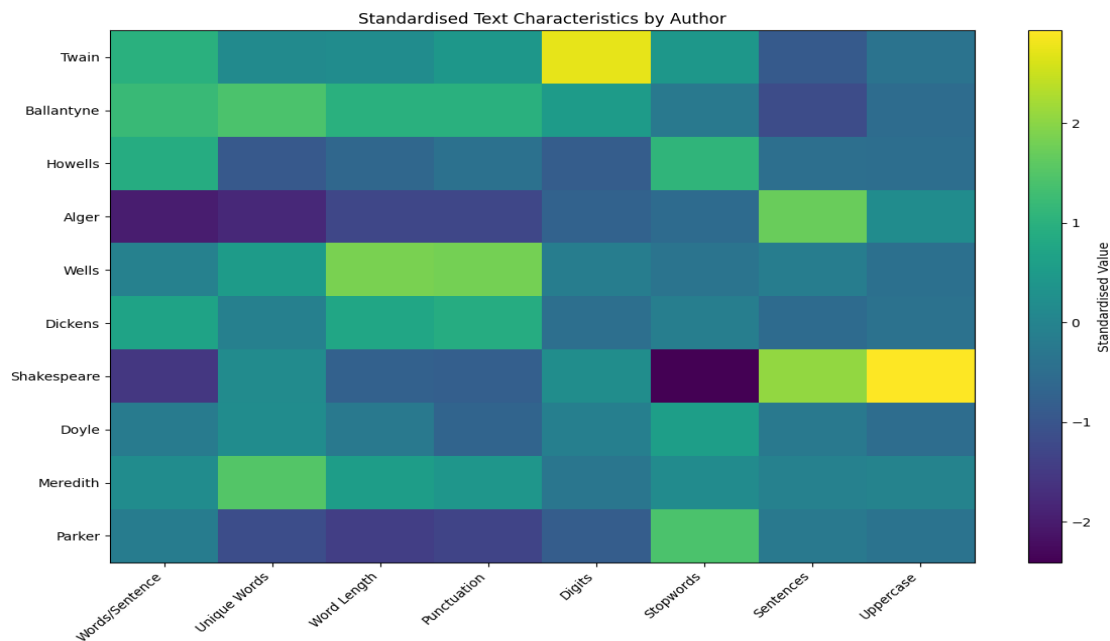








This analysis helped identify whether authors showed differences in vocabulary and general writing patterns. These differences provided a basis for investigating whether writing style could be used to distinguish between authors.



Overall, the EDA showed that the selected texts contained measurable differences in their linguistic characteristics. These findings supported the use of text-based features for the classification models.

Feature Selection and Engineering

The text had to be converted into numerical features before it could be used by the machine learning models. Three main approaches were considered.

Word-level TF-IDF was first used to represent important words in each passage. Common stopwords were removed to reduce unnecessary information.

Word n-grams were then tested by combining individual words with two-word sequences. This was intended to capture short phrases that may be associated with an author's writing style.

Finally, **character-level features** were developed using character 3-grams and 5-grams. Character patterns can capture features such as spelling, punctuation, word structure and common character combinations without relying only on individual words.

The character-level representation produced the strongest results during model testing and was therefore selected for the final model.

The feature engineering process therefore moved from general word patterns towards more detailed character-level patterns that could better represent differences in writing style.

Model Training

Several classification models were trained to determine which approach was most suitable for predicting the author of a passage.

Logistic Regression

Logistic Regression was used as the main baseline classification model. It was trained using the TF-IDF features extracted from the text. The model learns relationships between the text features and the different author classes.

Naive Bayes

Naive Bayes was also trained because it is commonly used for text classification. It provides a simple comparison with Logistic Regression and works well with high-dimensional text features.

Random Forest

Random Forest was tested as an alternative machine learning approach. It combines multiple decision trees to make predictions. However, its performance was considerably lower than the other models.

Word n-gram Logistic Regression

Word bigrams were added to the TF-IDF representation to capture short word sequences. A Logistic Regression model was then trained using the combined unigram and bigram features.

Character-Level Logistic Regression

A final Logistic Regression model was trained using **character 3-gram and 5-gram TF-IDF features**. This approach was intended to capture smaller patterns in the text that may reflect an author's writing style.

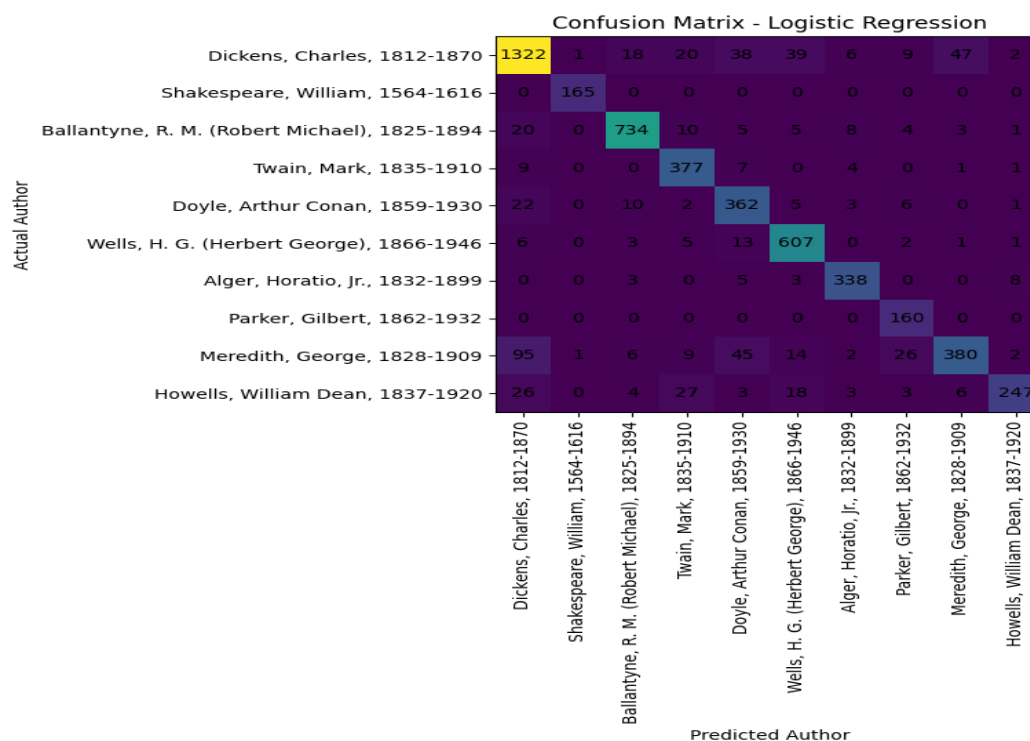
The models were trained using the training portion of the dataset and evaluated using the separate testing portion. Their results were compared using accuracy, precision, recall and F1-score.

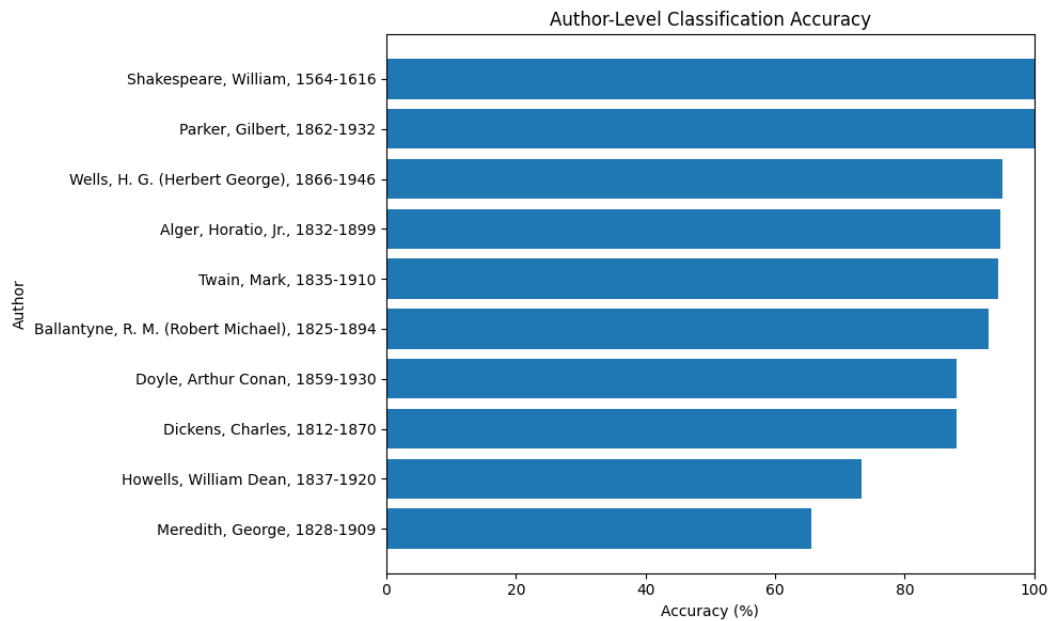
Model Evaluation and Interpretation

The models were evaluated using **accuracy, precision, recall and F1-score**. Confusion matrices and author-level accuracy were also used to understand where the models performed well and where they made errors.

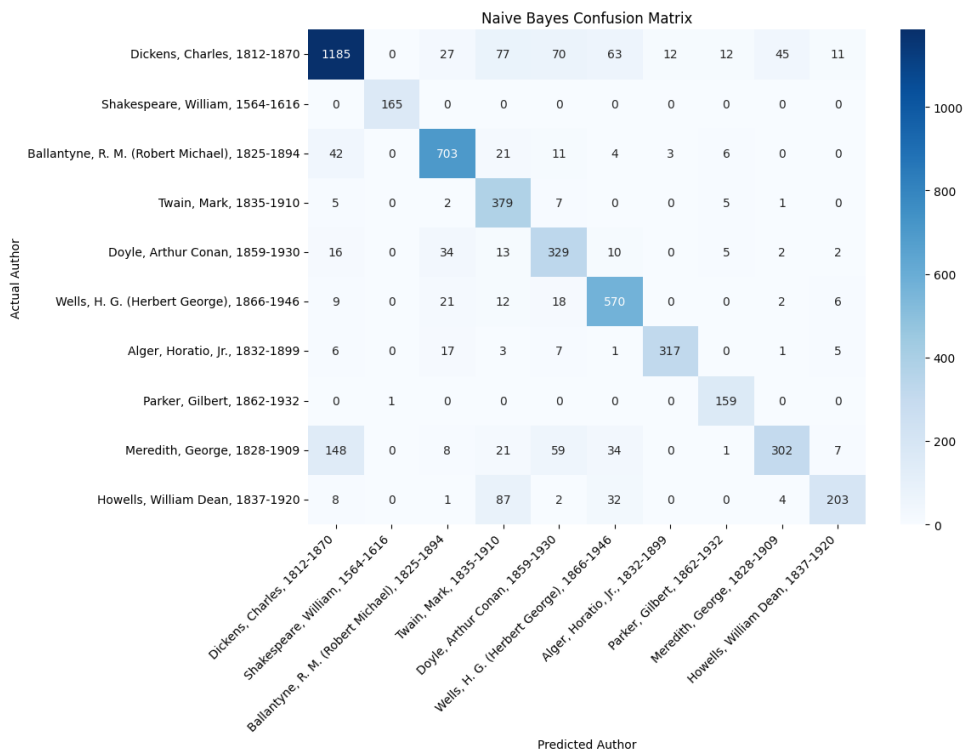
Logistic Regression

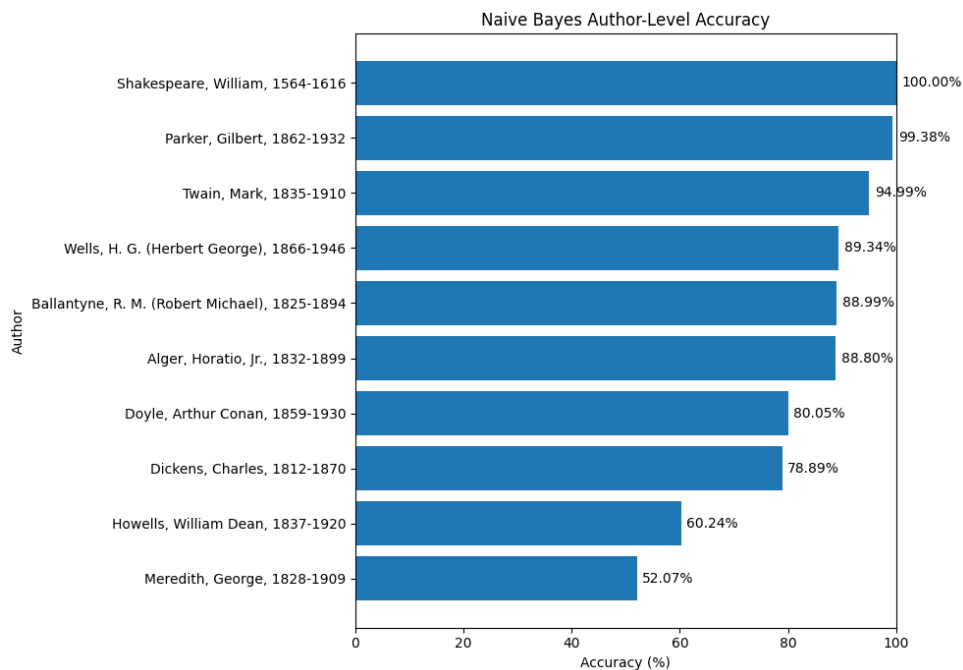
The original Logistic Regression model achieved an accuracy of **87.88%**. This provided a strong baseline for comparing the other approaches.





The confusion matrix shows which authors were correctly classified and which authors were confused with one another.





The results indicate that Naive Bayes was useful for text classification but was less effective than Logistic Regression for this dataset.

Random Forest and N-gram Models

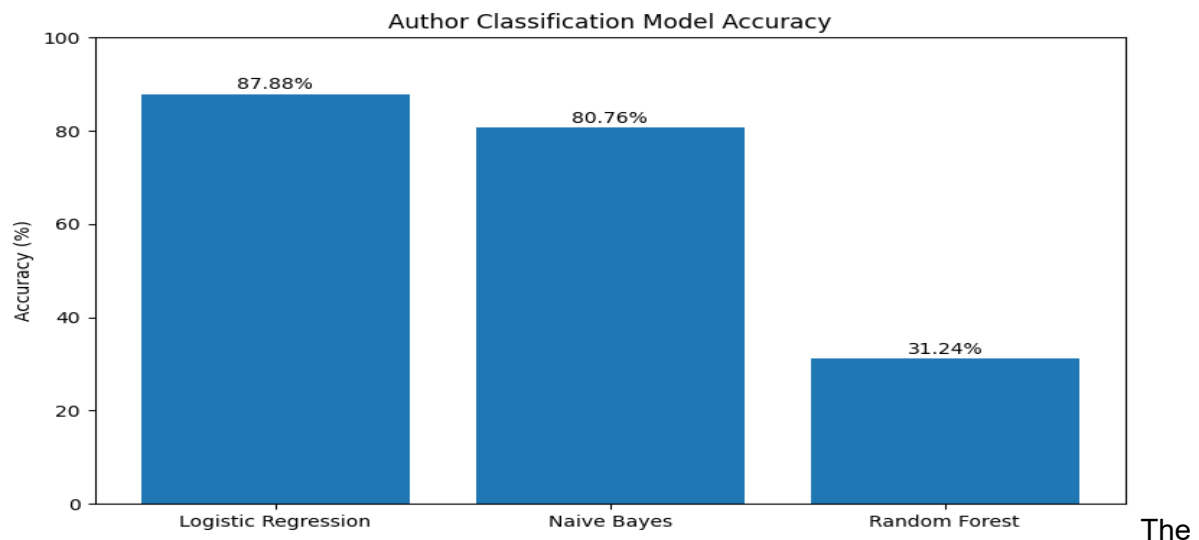
Random Forest performed poorly, with an accuracy of approximately **31.24%**. The word n-gram Logistic Regression model achieved approximately **86.35%**, which was also lower than the original Logistic Regression model.

These results showed that adding word-level bigrams did not improve the model and that Random Forest was not suitable for this feature representation.

Character-Level Model

The character-level Logistic Regression model produced the strongest test-set results:

Metric	Result
Accuracy	98.05%
Precision	98.08%
Recall	98.05%
F1-score	98.05%



improvement suggests that character-level patterns captured differences in writing style more effectively than the word-based features. The model was therefore selected for the final chatbot.

However, the high-test accuracy should be interpreted carefully because the dataset was split at passage level. Passages from the same book could potentially occur in both training and testing data, which may make the model appear more accurate than it would be when tested on completely unseen books.

Model Refinement and Retraining

After evaluating the initial models, further experiments were conducted to determine whether their performance could be improved.

Class Weighting

Class weights were introduced to account for differences in the number of passages available for each author. A weighted Logistic Regression model was trained so that authors with fewer passages received greater importance.

The weighted model achieved approximately **87.94% accuracy**, which was only a small improvement over the original Logistic Regression model. It was therefore not selected as the final model.

Word N-grams

Word bigrams were added to the TF-IDF features to capture short sequences of words. The resulting model achieved approximately **86.35% accuracy**, which was lower than the original Logistic Regression model.

Character-Level Features

Character 3-gram and 5-gram TF-IDF features were then tested. This produced a major improvement, with the Character-Level Logistic Regression model achieving **98.05% accuracy** on the test set.

Based on the comparison, the character-level model was selected as the final model because it provided the strongest overall performance.

The final model selected for the project was the **Character-Level Logistic Regression model**. It uses character 3-grams and 5-grams converted into TF-IDF features. This approach achieved the highest test-set performance, with an accuracy of **98.05%** and an F1-score of **98.05%**.

The trained model was then integrated into an interactive author prediction chatbot. The chatbot accepts a passage entered by the user, processes the text using the same character-level feature pipeline used during training, and sends the resulting features to the trained Logistic Regression model.

The chatbot returns the **three most likely authors** together with their predicted confidence scores. A confidence message is also displayed to indicate whether the prediction has high, moderate or low confidence.

```
... =====  
              FINAL AUTHOR PREDICTION BOT  
=====  
Model: Character-Level Logistic Regression  
Enter 'exit' to stop the bot.  
  
Enter a passage: 
```

Final chatbot process

1. User enters a passage of at least 10 words.
2. Text is converted to lowercase and prepared for character-level analysis.
3. Character 3-grams and 5-grams are generated.
4. Character TF-IDF features are created.
5. The trained Logistic Regression model predicts the author.
6. The three highest-probability authors are displayed.
7. A confidence level is provided for the prediction.

The final chatbot provides a practical demonstration of how the trained classification model can be used as an interactive tool for identifying likely authors from writing style.

```
*** =====
      FINAL AUTHOR PREDICTION BOT
=====
Model: Character-Level Logistic Regression
Enter 'exit' to stop the bot.

Enter a passage: Final chatbot process User enters a passage of at least 10 words. Text is converted to lowerca

Top predictions:
1. Doyle, Arthur Conan, 1859-1930 (75.84%)
2. Meredith, George, 1828-1909 (16.42%)
3. Howells, William Dean, 1837-1920 (3.01%)

✓ High confidence prediction.

Enter a passage: There are moments in life when a man discovers that the things he believed to be important wer

Top predictions:
1. Doyle, Arthur Conan, 1859-1930 (51.48%)
2. Parker, Gilbert, 1862-1932 (17.85%)
3. Shakespeare, William, 1564-1616 (11.50%)

⚠ Moderate confidence prediction.

Enter a passage: It was a dreadful thing to see, and I wished I had never come there. I had always been a carel

Top predictions:
1. Doyle, Arthur Conan, 1859-1930 (91.40%)
2. Howells, William Dean, 1837-1920 (3.72%)
3. Shakespeare, William, 1564-1616 (1.98%)

✓ High confidence prediction.

Enter a passage: 
```

Conclusion and Recommendations

The project successfully developed an author prediction chatbot using the SPGC dataset. Several models were compared, with **Character-Level Logistic Regression** achieving the best performance at **98.05% accuracy** and **98.05% F1-score**.

The final chatbot accepts a text passage and returns the three most likely authors with confidence scores.

Recommendations

- Use **book-level train/test splitting** to reduce possible data leakage.
- Test the model on completely unseen books.
- Add more authors and books.
- Include an **unknown author** option for texts outside the trained authors.

Overall, the final model provides a strong prototype for an interactive authorship prediction tool.

Reference List

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Sapkota, U., Bethard, S., Montes, M. and Solorio, T. (2015) 'Not all character n-grams are created equal: A study in authorship attribution', *Proceedings of the 2015 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pp. 93–102.

Dataset

Gerlach, M. and Font-Clos, F. (2018) *Standardized Project Gutenberg Corpus (SPGC-2018-07-18)*. Zenodo.